Hasib ur Rahman

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LinkedIn | ResearchGate | GitHub

EDUCATION

B.Sc. in Software Engineering, Major in Data Science July 2023 – July 2027 (Expected)
Daffodil International University, Dhaka, Bangladesh
CGPA: 3.70/4.00

Thesis (Ongoing): RainLite-YOLO: A Physics-Guided Feature-Space Progressive Deraining and Lightweight Object Detection Network for Autonomous Vehicles in Rainy Weather

Field(s) of Study: Machine Learning, Deep Learning, Machine Learning Driven Data Analysis, Artificial Intelligence, Database Management, Software Engineering, Data Structures & Algorithms, Object-Oriented Programming, Statistics, Linear Algebra, Computer Networks, Data Communication, Economics, Programming & Problem Solving

RESEARCH INTEREST

- Medical Image Analysis
- Digital Image Processing
- 3D Computer Vision
- Explainable AI (XAI)
- Computer Aided Diagnosis

WORK EXPERIENCE

Research Intern — ELITE Research Lab LLC October 2026 – Present — Dhaka, Bangladesh

- Designing and optimizing a **lightweight CNN** for **cross-magnification whole-slide image analysis** in **prostate cancer diagnosis** using **Python** and **PyTorch**.
- Performing **literature reviews** and contributing to **manuscript preparation** for future publication.

Machine Learning Engineer — Fosholer Chikitsha March 2025 – June 2026 — Dhaka, Bangladesh

- Developed and deployed end-to-end **machine learning pipelines** using **CNNs** and **Vision Transformers (ViT)** for automated crop disease classification and prevention guidance.
- Optimized **model efficiency**, reliability, and interpretability for high performance in low-connectivity and low-image-quality environments.
- Engineered **real-time disease detection systems** and field guidance solutions tailored for agricultural applications.

RESEARCH EXPERIENCE

1. **Hasib ur Rahman**, Imtiaz Ibna Kamal, Md Abdullah Al Maruf, Dhroubo Jyoti Mondal. **"DiaStackBD: An Explainable Machine Learning Framework for Type 2 Diabetes Prediction in Bangladesh with Lightweight Web Deployment."** *5th IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON).* **2026.** [**Presented**] [Certificate] [Paper]

2. **Hasib ur Rahman**, Md Babul Hossain, Imtiaz Ibna Kamal, Imranul Islam Shihab, Izaz Ahmmed Tuhin, Md Rajib Mia. **"Knowledge Distillation from Multi-Teacher CNNs for Explainable and Lightweight Brain Tumor MRI Classification in Low-Resource Settings."** *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT).* **2026.** [**Under Review**]

3. **Hasib ur Rahman**, Imtiaz Ibna Kamal, Rashidul Hassan Hridoy, Imranul Islam Shihab, Dhroubo Jyoti Mondal. **"A Lightweight and Deployable Tool for Real-Time Polyp Segmentation in Colonoscopy Images Using Deep Learning."** *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT).* **2026.** [**Under Review**]

4. **Hasib ur Rahman**, Imtiaz Ibna Kamal, Mahbub Hassan Hira, Dhroubo Jyoti Mondal, Md Abdullah Al Maruf. **"Automated Breast Cancer Detection in Mammography Using Preprocessed Transfer Learning Architectures and Feature Fusion."** *7th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT).* **2026.** [**Under Review**]

SKILLS

Research Field:	Statistics, Machine Learning, Image Processing, Image Analysis, Image Segmentation, Computer Vision, Jupyter Notebook, LaTeX, Zotero
Programming Languages:	Python, R, Java, C++, JavaScript
Frameworks & Tools:	PyTorch, Scikit-learn, FastAPI, HTML, CSS, Git/GitHub, Docker, GitLab
Platforms:	Kaggle, Google Colab

PROJECTS

1. **NeuroScan: Lightweight Explainable Brain Tumor MRI Classifier** Github — Live App
   Tech: Python, PyTorch, CNN, Knowledge Distillation, ShuffleNetV2, Grad-CAM++, LIME, FastAPI

   Built a **lightweight 5-block CNN** using **knowledge distillation** from a **ShuffleNetV2** teacher model for **4-class MRI classification**, achieving **93.38% test accuracy**. Reduced model size by **2.60**× and boosted inference speed by ∼**9.65**×; integrated real-time predictions with **Grad-CAM++** and **LIME** visual explainability in a **FastAPI** web app.

2. **Accurate Brain Tumor Detection using Swin Transformer** Github
   Tech: Python, PyTorch, Swin Transformer, W-MSA, AMP, Cosine Annealing

   Developed a multi-stage **Swin Transformer** for **4-class MRI brain tumor classification**, achieving **93% accuracy**. Leveraged **shifted window self-attention (W-MSA)** and **patch merging** with **Automatic Mixed Precision (AMP)** and **Cosine Annealing** optimization.

3. **Breast Cancer Detection using Machine Learning** Github
   Tech: Python, Scikit-learn, Logistic Regression, Random Forest, SVM

   Engineered an **ML pipeline** for **binary breast tumor classification** featuring **EDA**, **feature scaling**, and **feature selection**. Tuned **Logistic Regression**, **Random Forest**, and **SVM** models to maximize accuracy and recall.

AWARDS & ACHIEVEMENTS

- **11th Rank, Datathon at AI Dev Fest 2026**, Daffodil International University – Organized by DIU CPC under the Department of CSE *(October 2026)*
- **Brilliant Academic Result Scholarship**, Daffodil International University – Awarded across multiple semesters, waiving tuition fees by up to 30% *(2026)*
- **Finalist, IEEE ICADHI 2026** – Presented a Physics-Guided Temporal ECG Signal Analysis System for clustering cardiac conditions and real-time abnormality alerts *(June 2026)*
- **5th National Data Science Summit 2026 (Data Hackathon)** – Built an ML-based safe navigation tool for merchant ships *(June 2026)*
- **Finalist, Undergraduate Startup Challenge 2025** (EBL Women Network & Eastern Bank PLC) – Presented business model and growth strategy for Fosholer Chikitsha *(January 2026)*
- **Pupil Rank, Codeforces** – Highest rating 1296; solved nearly 400 competitive programming problems *(2025)*
- **Top 8%, CodeTrap Programming Contest Spring 2025**, Daffodil International University – Solved 5/8 DSA-based programming problems *(2025)*
- **Finalist, Project Showcasing Segment, Prime Bank Presents EWU National RoboFest 2024** – Organized by the East West University Robotics Club; presented a self-built *Smart Lab Security System (2024)*
- **Top 5% Finalist, Googling Contest Spring 2024**, Daffodil International University *(February 2024)*